

ENGINEERING PHYSICS

Complete Exam-Ready Answer Book

Anna University | B.E./B.Tech First Year

Strictly as per Latest Syllabus & Valuation Pattern



Prepared for Maximum Marks (13-Mark Questions)

Simple Language | Complete Derivations | Neat Diagrams

All Formulae | Key Points | Exam Tips

PART A

Question 1: How many crystal systems are there? Classify each of them.

[13 Marks | Anna University Pattern]

1. INTRODUCTION TO CRYSTAL SYSTEMS

A crystal is a solid in which atoms are arranged in a regular, repeating pattern called a crystal lattice. The entire crystal structure can be built by repeating a small unit called the **Unit Cell**. Based on the shape and symmetry of unit cells, all crystals are classified into **7 Crystal Systems**.

2. THE SEVEN CRYSTAL SYSTEMS

The seven crystal systems are distinguished by three edge lengths **a, b, c** (lattice parameters) and three angles α, β, γ between the axes.

S.No	Crystal System	Axial Lengths	Axial Angles	Example
1	Cubic (Isometric)	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	NaCl, Cu, Fe
2	Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	TiO ₂ , SnO ₂
3	Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	BaSO ₄ , KNO ₃
4	Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$	CaSO ₄ ·2H ₂ O
5	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	CuSO ₄ ·5H ₂ O
6	Trigonal (Rhombohedral)	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	Calcite, Quartz
7	Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	Zn, Mg, Graphite

3. THE 14 BRAVAIS LATTICES

Each crystal system can have different types of unit cell arrangements. Auguste Bravais showed that there are exactly **14 unique lattice types** (Bravais Lattices) in total across all 7 systems.

Crystal System	No. of Bravais Lattices	Types
Cubic	3	SC, BCC, FCC
Tetragonal	2	Simple, Body-Centred
Orthorhombic	4	Simple, BC, FC, End-Centred
Monoclinic	2	Simple, End-Centred
Triclinic	1	Simple
Trigonal	1	Simple
Hexagonal	1	Simple
TOTAL	14	—

4. DETAILED DESCRIPTION OF EACH CRYSTAL SYSTEM

(i) Cubic System

All three axes equal; all angles 90° . Most symmetric system. Contains SC (Simple Cubic), BCC (Body Centred Cubic) and FCC (Face Centred Cubic). Example: NaCl (rock salt), Copper, Iron, Diamond.

(ii) Tetragonal System

Two axes equal, one different; all angles 90° . Like a stretched cube along one axis. Has Simple and Body-Centred. Example: White tin (β -Sn), Indium, TiO_2 (Rutile).

(iii) Orthorhombic System

All three axes unequal; all angles 90° . Like a rectangular box. Has 4 Bravais lattices. Example: Sulphur, Topaz, Barium Sulphate.

(iv) Monoclinic System

All three axes unequal; two angles $= 90^\circ$ but $\beta \neq 90^\circ$. Has 2 lattices. Example: Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), Borax, Monoclinic Sulphur.

(v) Triclinic System

All three axes unequal; all angles unequal and none equal to 90° . Least symmetric. Only 1 Bravais lattice. Example: Copper Sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), Potassium Dichromate.

(vi) Trigonal / Rhombohedral System

All three axes equal; all angles equal but not 90° . Like a cube that has been tilted. Example: Calcite, Bismuth, Arsenic, Quartz (one form).

(vii) Hexagonal System

Two axes equal, third different; two angles $= 90^\circ$, one angle $= 120^\circ$. Unit cell looks like a hexagonal prism. Example: Zinc, Magnesium, Graphite, Beryllium.

5. DIAGRAM — The 7 Crystal Systems

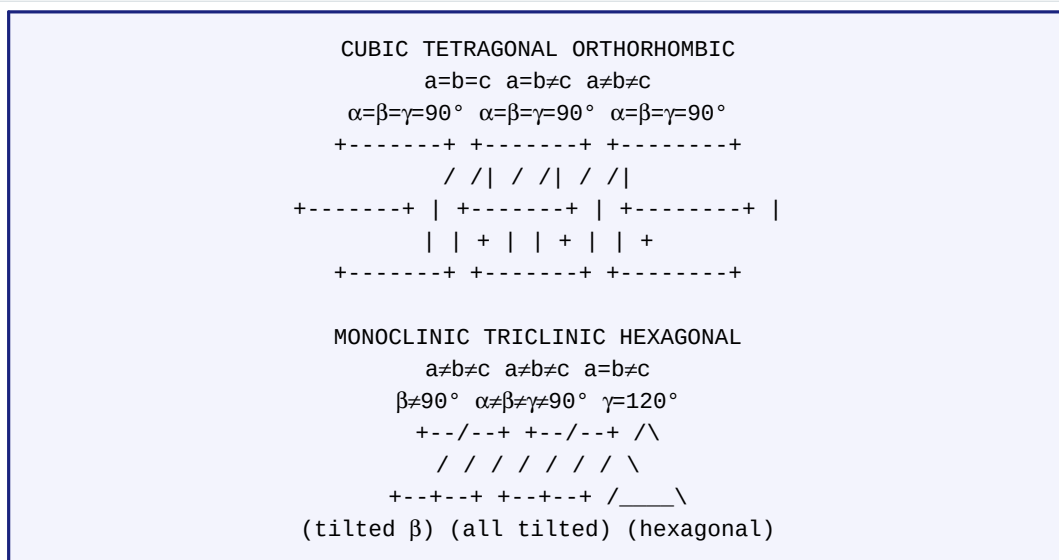


Fig: 7 Crystal Systems — Lattice Parameter Comparison

6. KEY POINTS FOR EXAMINATION

- There are **7 crystal systems** and **14 Bravais lattices** in total.
- Cubic system has the **highest symmetry**; Triclinic has the **lowest symmetry**.

- Axial lengths: a , b , c ; Axial angles: α (between b & c), β (between a & c), γ (between a & b).
- FCC and BCC are important for engineering metals like iron, copper, aluminium.
- The **Hexagonal system** has $\gamma = 120^\circ$ — this is a unique and frequently asked point.

★ **EXAM TIP:** Always write the table of 7 crystal systems with axial lengths, angles and examples — this alone fetches 6–7 marks. Add the 14 Bravais lattices table for extra marks.

PART A

Question 2: Explain with a neat diagram the Miller Indices of (020), (-100) and (202) planes.

[13 Marks | Anna University Pattern]

1. WHAT ARE MILLER INDICES?

Miller Indices are a set of three integers (h, k, l) used to describe the orientation of crystal planes in a crystal lattice. They were introduced by **W.H. Miller** in 1839.

2. HOW TO FIND MILLER INDICES — STEP-BY-STEP PROCEDURE

Step 1: Find where the plane cuts the X, Y and Z axes (intercepts).

Step 2: Write the intercepts in terms of lattice parameters (a, b, c).

Step 3: Take the RECIPROCALs of the intercepts.

Step 4: Clear fractions by multiplying by the LCM.

Step 5: Enclose the result in parentheses (h k l) — this is the Miller Index.

Step 6: A negative intercept is shown with a bar over the number (e.g., -1 is written as $\bar{1}$).

3. PLANE (020)

Intercepts: $X = \infty$, $Y = 1/2$, $Z = \infty$

Reciprocals: $1/\infty = 0$, $1/(1/2) = 2$, $1/\infty = 0$

Miller Index $\rightarrow$ **(0 2 0)**

This plane is PARALLEL to the X and Z axes and cuts the Y-axis at $b/2$.

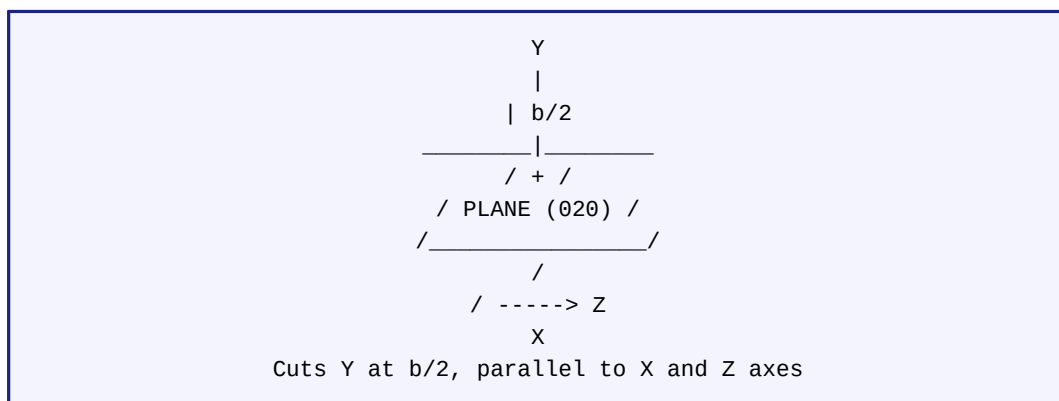


Fig: Miller Plane (020)

4. PLANE (-100) — Negative Index Plane

Intercepts: $X = -1$, $Y = \infty$, $Z = \infty$

Reciprocals: $1/(-1) = -1$, 0 , 0

Miller Index $\rightarrow$ **($\bar{1}$ 0 0)** written as ($\bar{1}$ 0 0)

This plane is PARALLEL to the Y and Z axes and cuts the NEGATIVE X-axis at $-a$.

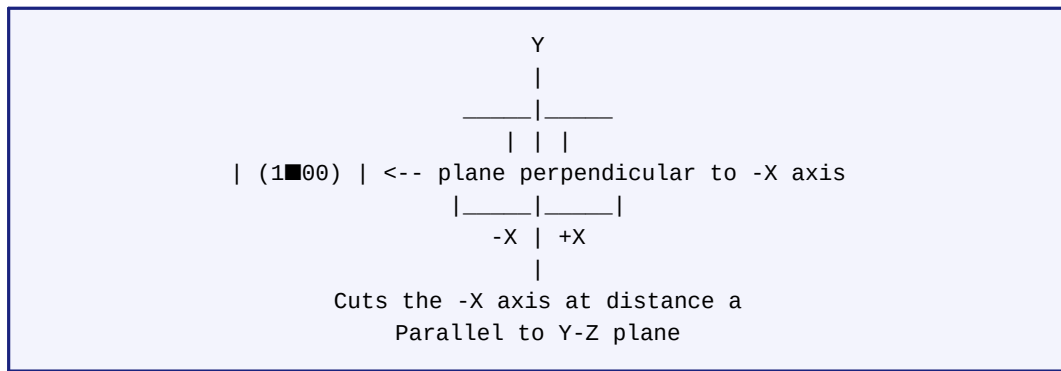


Fig: Miller Plane $(-100) = (1\bar{0}0)$

5. PLANE (202)

Intercepts: $X = 1, Y = \infty, Z = 1$

Reciprocals: $1/1 = 1, 1/\infty = 0, 1/1 = 1 \rightarrow$ already integers

Hmm! But (202) means $h=2, k=0, l=2$. So reciprocals were $1/2, 1/\infty, 1/2$

Intercepts: $X = a/2, Y = \infty, Z = c/2$

Reciprocals: 2, 0, 2 $\rightarrow$ Miller Index = **(2 0 2)**

This plane cuts the X-axis at $a/2$ and Z-axis at $c/2$; parallel to Y-axis.

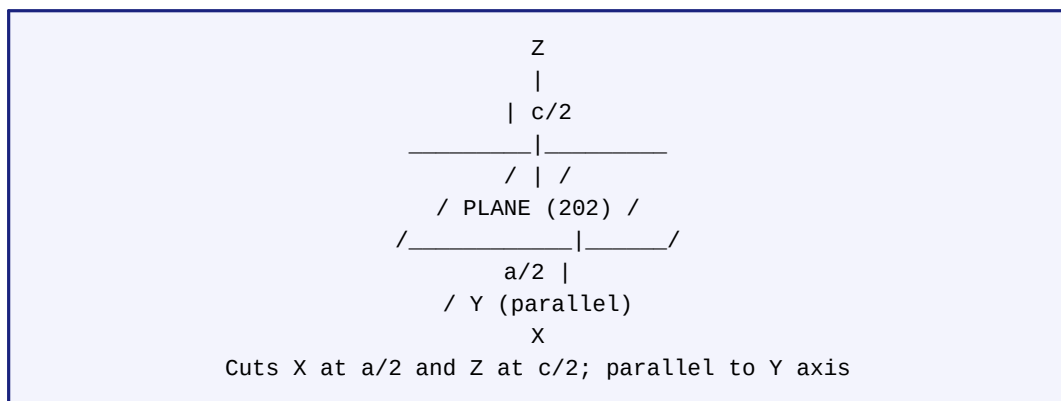


Fig: Miller Plane (202)

6. IMPORTANT PROPERTIES OF MILLER INDICES

- If a plane is parallel to an axis, the intercept is ∞ and the Miller index is 0.
- Negative intercept $\rightarrow$ negative index $\rightarrow$ written with a bar on top.
- (hkl) denotes a specific plane; $\{hkl\}$ denotes all equivalent planes (family).
- $[hkl]$ denotes a direction; $\langle hkl \rangle$ denotes a family of directions.
- For a cubic crystal, the plane (hkl) is perpendicular to the direction $[hkl]$.

★ **EXAM TIP:** In exams, always show the step-by-step procedure (intercepts $\rightarrow$ reciprocals $\rightarrow$ clear fractions) and draw the diagram. Writing both the procedure AND the diagram is necessary for full marks.

PART A

Question 3: Explain the NaCl structure with a neat diagram.

[13 Marks | Anna University Pattern]

1. INTRODUCTION

Sodium Chloride (NaCl), commonly known as table salt, is an ionic compound that forms a very important crystal structure. The **NaCl structure** is the prototype for many ionic crystals.

2. CRYSTAL STRUCTURE OF NaCl

The NaCl structure consists of two interpenetrating FCC (Face Centred Cubic) sublattices:

- One FCC sublattice is formed by Na⁺ ions.
- The other FCC sublattice is formed by Cl⁻ ions.
- The Cl⁻ FCC lattice is displaced by $a/2$ along one axis from the Na⁺ FCC lattice.

3. IMPORTANT STRUCTURAL DETAILS

Property	Value
Crystal System	Cubic
Bravais Lattice	Face Centred Cubic (FCC)
Coordination Number (Na ⁺)	6 (surrounded by 6 Cl ⁻)
Coordination Number (Cl ⁻)	6 (surrounded by 6 Na ⁺)
No. of Na ⁺ per unit cell	4
No. of Cl ⁻ per unit cell	4
Total ions per unit cell	8 (4 NaCl formula units)
Lattice Parameter (a)	5.64 Å
Nearest neighbour distance	$a/2 = 2.82 \text{ Å}$
Next nearest distance	$a/\sqrt{2} = 3.99 \text{ Å}$

4. POSITION OF IONS IN UNIT CELL

Na⁺ ion positions:

- Body centre: $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) \times 1 = 1$ ion
- Edge centres (12 edges): $(\frac{1}{4}, 0, 0)$ type $\times 12 \times (\frac{1}{4}) = 3$ ions
- Total Na⁺ = $1 + 3 = 4$ ions per unit cell

Cl⁻ ion positions:

- Corners (8 corners): $(0,0,0)$ type $\times 8 \times (\frac{1}{8}) = 1$ ion
- Face centres (6 faces): $(\frac{1}{2}, \frac{1}{2}, 0)$ type $\times 6 \times (\frac{1}{2}) = 3$ ions
- Total Cl⁻ = $1 + 3 = 4$ ions per unit cell

5. NEAT DIAGRAM OF NaCl STRUCTURE

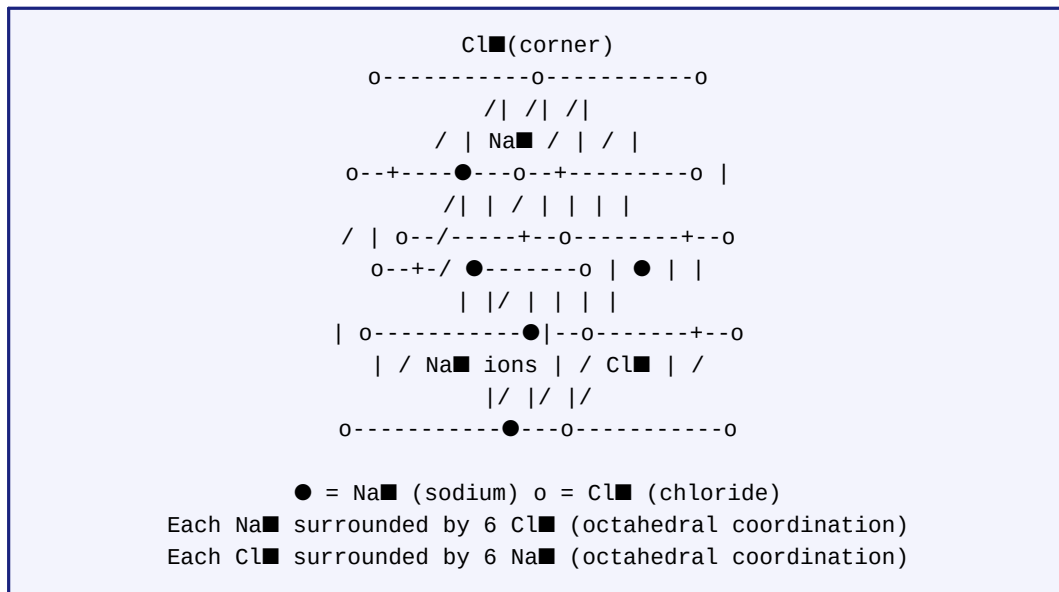


Fig: Crystal Structure of NaCl (Rock Salt Structure)

6. PACKING EFFICIENCY

In NaCl, the packing is not purely hard-sphere; it is ionic. Radius ratio r/R (Na■/Cl■) = $0.102/0.181 \text{ nm} = 0.564$, which falls in the range 0.414–0.732, confirming **octahedral coordination** (CN = 6).

7. APPLICATIONS AND EXAMPLES

- NaCl structure is adopted by: NaCl, KCl, MgO, CaO, FeO, AgBr, MnO
- Used as a standard for X-ray diffraction calibration.
- MgO (Magnesia) with NaCl structure is used in refractory materials.

★ **EXAM TIP:** Draw a clear 3D diagram showing both Na■ and Cl■ ions. Mention coordination number = 6, number of atoms per unit cell = 4 formula units, and lattice parameter = $5.64 \text{ \AA}$ — these are the mark-fetching key values.

PART A

Question 4: Distinguish Frenkel and Schottky defects and explain the terms.

[13 Marks | Anna University Pattern]

1. INTRODUCTION TO CRYSTAL DEFECTS

A perfect crystal exists only at 0 K. At all other temperatures, crystals contain **defects** — deviations from the perfect lattice arrangement. Point defects are the simplest type, occurring at specific lattice sites. The two most important point defects are **Frenkel Defect** and **Schottky Defect**.

2. FRENKEL DEFECT

Definition:

When an atom or ion is displaced from its regular lattice site and moves to an interstitial (gap) position in the same crystal, leaving behind a vacancy, the pair (vacancy + interstitial) is called a **Frenkel Defect**.

Characteristics:

- The displaced atom sits between regular lattice sites (interstitial position).
- The crystal as a whole remains electrically neutral.
- No change in volume or density of the crystal.
- Commonly occurs in ionic crystals where cations are much smaller than anions.
- Examples: AgBr, AgCl, AgI, ZnS — silver and zinc halides.
- In AgBr: Ag⁺ ions (small) jump to interstitial positions; Br⁻ stays in place.

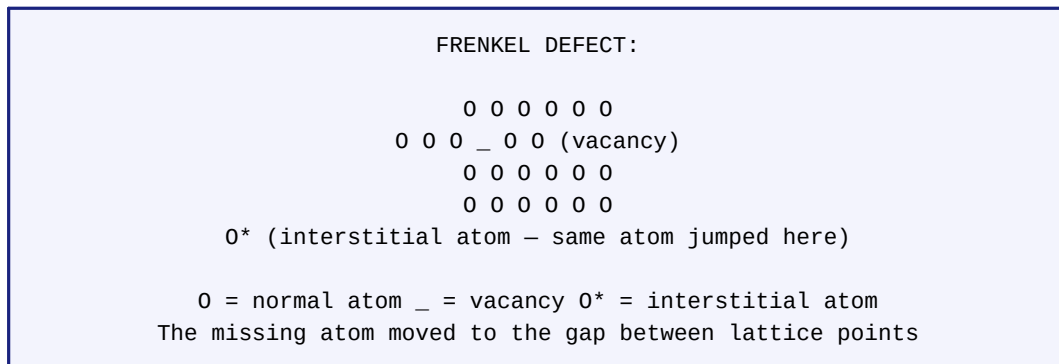


Fig: Frenkel Defect — Atom displaced to interstitial position

3. SCHOTTKY DEFECT

Definition:

When an atom or ion is completely removed from a regular lattice site and migrates to the surface of the crystal, the resulting vacancy is called a **Schottky Defect**. In ionic crystals, equal numbers of positive and negative ion vacancies are created (to maintain electrical neutrality).

Characteristics:

- The atom permanently leaves the lattice and goes to the crystal surface.
- Results in a PAIR of vacancies (cation vacancy + anion vacancy) in ionic crystals.
- Density of the crystal DECREASES (volume increases slightly).
- Occurs in crystals where cation and anion sizes are nearly equal.
- Examples: NaCl, KCl, KBr, CsCl — alkali metal halides.

- In NaCl: one Na■ vacancy + one Cl■ vacancy formed together.

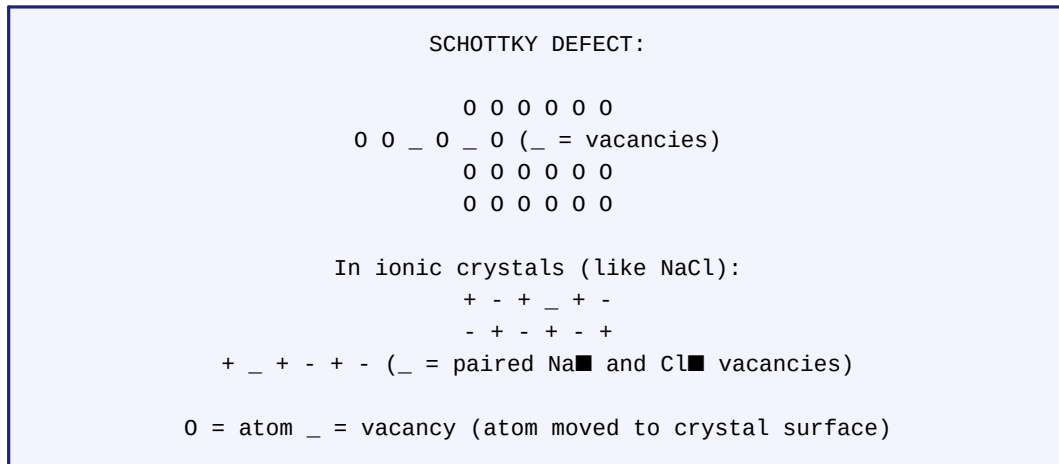


Fig: Schottky Defect — Equal cation and anion vacancies

4. COMPARISON TABLE — FRENKEL vs SCHOTTKY DEFECTS

Property	Frenkel Defect	Schottky Defect
Definition	Atom displaced to interstitial position	Atom removed to crystal surface
Vacancies	1 vacancy + 1 interstitial atom (pair)	Only vacancies, no interstitial
Density	Remains unchanged	Decreases
Volume	Remains unchanged	Slightly increases
Ion size	Large difference between cation and anion size	Similar size of cation and anion
Electrical neutrality	Maintained	Maintained (equal pairs in ionic)
Dielectric constant	Increases	No significant change
Examples	AgBr, AgCl, ZnS	NaCl, KCl, KBr
Temperature dependence	Increases with temperature	Increases with temperature

5. ENERGY OF FORMATION

For Schottky defect: $n = N \times \exp(-E_s / 2kT)$

Where: n = number of Schottky defect pairs, N = total number of atoms, E_s = energy to form one Schottky pair, k = Boltzmann's constant, T = temperature in Kelvin

For Frenkel defect: $n = \sqrt{(N \times N_i) \times \exp(-E_f / 2kT)}$

Where: N_i = number of interstitial sites, E_f = energy to form one Frenkel defect, other symbols same as above

★ **EXAM TIP: The comparison table is very important — draw it clearly. Also mention that NaCl shows Schottky defect and AgBr shows Frenkel defect. Write both formulas for 2 extra marks.**

PART A

Question 5: Define Bragg's Law and explain the terms.

[13 Marks | Anna University Pattern]

1. INTRODUCTION

When X-rays fall on a crystal, they are reflected (diffracted) by the planes of atoms. **Bragg's Law** gives the condition for constructive interference of X-rays reflected from parallel crystal planes. This is fundamental to **X-ray crystallography** — the science of determining crystal structure.

2. BRAGG'S LAW — STATEMENT

Bragg's Law states: 'When X-rays are incident on a crystal, the reflected rays from successive parallel crystal planes will undergo constructive interference only when the path difference between them equals an integral multiple of the wavelength.'

3. DERIVATION OF BRAGG'S LAW

Consider a set of parallel crystal planes separated by an interplanar spacing **d**. Let X-rays of wavelength λ be incident on these planes at a glancing angle θ (angle between the incident ray and the crystal plane).

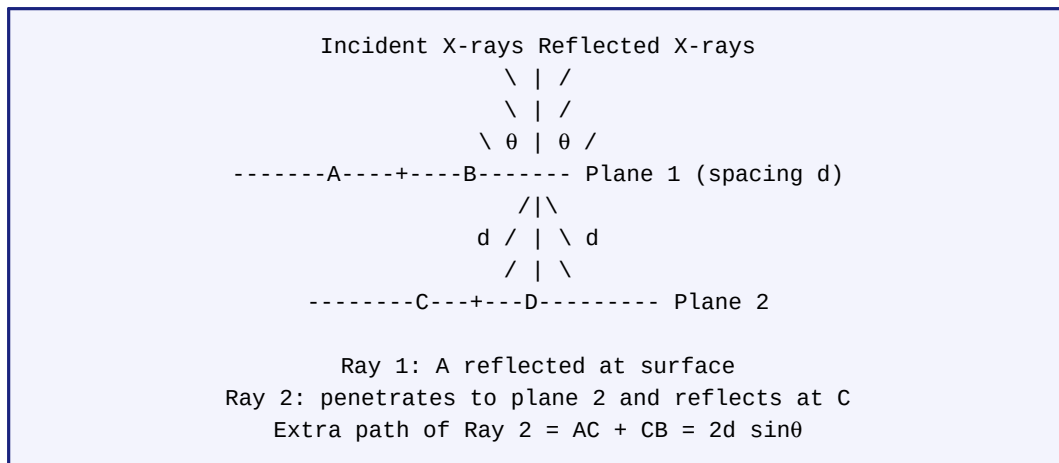


Fig: Bragg's Law Derivation — X-ray reflection from crystal planes

From the diagram:

The extra path travelled by Ray 2 compared to Ray 1 = AC + CB

Since the geometry shows: AC = CB = d sin θ

$$\therefore \text{Path difference} = AC + CB = 2d \sin\theta$$

For **constructive interference** (bright reflected beam), path difference = $n\lambda$ (integer multiple of wavelength):

$$\therefore 2d \sin\theta = n\lambda$$

4. EXPLANATION OF TERMS IN BRAGG'S LAW: $2d \sin\theta = n\lambda$

Symbol	Meaning	Unit
n	Order of diffraction (n = 1, 2, 3, ...) Integer; n=1 is first-order	No unit (dimensionless)

d	Interplanar spacing — perpendicular distance between adjacent parallel planes	Ångström (Å) or nm
θ	Glancing angle — angle between incident X-ray and crystal plane surface	Degrees (°)
λ	Wavelength of incident X-rays	Ångström (Å) or nm

5. PHYSICAL SIGNIFICANCE

- **d** can be calculated if θ and λ are known: $d = n\lambda / (2 \sin\theta)$
- λ can be found if d and θ are known: useful for X-ray spectroscopy.
- Maximum diffraction occurs when $\theta = 90^\circ \rightarrow 2d = n\lambda \rightarrow \text{minimum } \lambda = 2d$ (for $n=1$).
- Bragg diffraction is the basis of XRD (X-Ray Diffraction) technique.

6. APPLICATIONS OF BRAGG'S LAW

- Determination of crystal structure (interplanar spacings, lattice parameters).
- Identification of unknown crystalline materials.
- Study of alloys, minerals, and polymers.
- Used in Bragg spectrometer to measure X-ray wavelengths.
- Medical: bone structure analysis.

★ **EXAM TIP: The derivation with diagram is mandatory. Write clearly: 'Path difference = $2d \sin\theta$, for constructive interference = $n\lambda$, therefore $2d \sin\theta = n\lambda$ '. Explain all four symbols individually.**

PART A

Question 6: Define Rigid Body.

[13 Marks | Anna University Pattern]

1. DEFINITION OF RIGID BODY

A **Rigid Body** is defined as a body in which the distance between any two particles remains **constant and unchanged**, regardless of the external forces applied to it. In other words, a rigid body does not deform, bend, compress or stretch under any applied force.

2. SIMPLE EXPLANATION

Imagine two points A and B inside a body. If the body is rigid, then the distance AB always stays the same no matter how we push, pull, or rotate the body. The shape and size of a rigid body never change.

3. TYPES OF MOTION IN A RIGID BODY

- **Translational Motion:** Every particle in the body moves in parallel paths with the same velocity and acceleration.
- **Rotational Motion:** The body rotates about a fixed axis. Each particle moves in a circle about the axis.
- **Combined Motion:** Both translation and rotation occurring together (e.g., a rolling wheel).

4. EXAMPLES

- A steel rod under normal conditions — does not compress under moderate force.
- A spinning top — rigid body undergoing rotational motion.
- A football kicked in the air — rigid body undergoing translational motion.
- A car wheel rolling — rigid body with combined motion.

5. DEGREES OF FREEDOM OF A RIGID BODY

A rigid body in 3D space has **6 degrees of freedom**:

- 3 translational: motion along X, Y, Z axes.
- 3 rotational: rotation about X, Y, Z axes.

6. NOTE — IDEAL vs REAL BODIES

A perfectly rigid body is an idealization. No real object is perfectly rigid — all materials deform slightly. However, for most engineering problems, we assume bodies are rigid when deformations are negligibly small.

★ **EXAM TIP:** For a 13-mark question, expand with types of motion, degrees of freedom, examples, and comparison between rigid and non-rigid bodies.

PART A

Question 7: Briefly explain Centre of Mass.

[13 Marks | Anna University Pattern]

1. DEFINITION

The **Centre of Mass (CoM)** of a body or a system of bodies is the unique point where the entire mass of the system may be assumed to be concentrated for the purpose of analyzing translational motion. It is the point where the resultant of all gravitational forces (weight) acts.

2. FORMULA FOR CENTRE OF MASS

For a system of n particles with masses $m_1, m_2, \dots, m_n$ at positions $(x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_n, y_n, z_n)$:

$$X_{cm} = (m_1x_1 + m_2x_2 + \dots + m_nx_n) / (m_1 + m_2 + \dots + m_n) = \Sigma m_i x_i / M$$

$$Y_{cm} = \Sigma m_i y_i / M$$

$$Z_{cm} = \Sigma m_i z_i / M$$

Where M = total mass = Σm_i

3. CENTRE OF MASS OF UNIFORM BODIES

Body	Location of Centre of Mass
Uniform rod	Geometric centre (midpoint)
Uniform circular disc	Centre of the disc
Sphere (uniform)	Centre of the sphere
Rectangle / Square	Intersection of diagonals
Triangle	Centroid (1/3 height from base)
Ring	Centre of the ring (may be empty space)

4. DIAGRAM

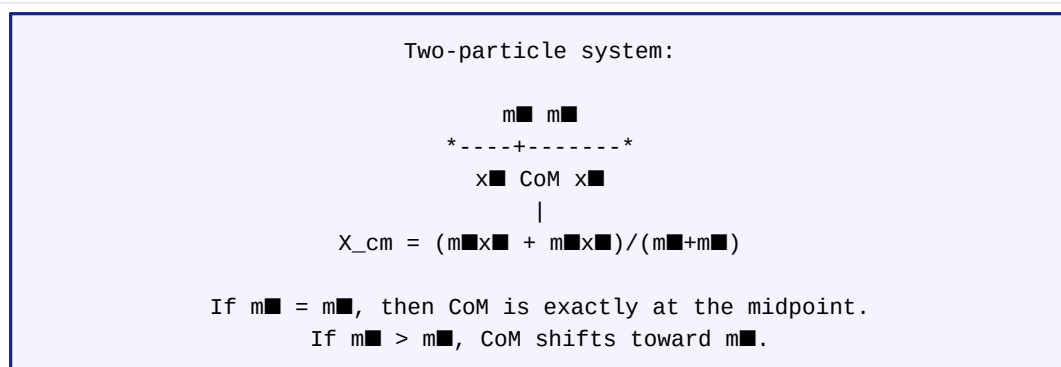


Fig: Centre of Mass of a Two-Particle System

5. KEY PROPERTIES

- The CoM of a symmetric, uniform body lies on the axis of symmetry.
- External forces produce translational acceleration of the CoM: $F = M \times a_{cm}$

- If no external force, the CoM moves with constant velocity (Newton's 1st Law).
- The CoM may lie outside the physical body (e.g., a ring or hollow sphere).
- For a system of bodies, CoM is the mass-weighted average position.

★ **EXAM TIP:** Write the formula, explain all symbols, draw the two-particle diagram, and give examples of CoM positions. Mention that CoM can lie outside the body.

PART A

Question 8: Briefly explain Moment of Inertia of a circular rod.

[13 Marks | Anna University Pattern]

1. DEFINITION OF MOMENT OF INERTIA

The **Moment of Inertia (I)** is the rotational analog of mass. It measures the resistance of a body to angular acceleration. It depends on both the mass of the body and how that mass is distributed about the axis of rotation.

$$I = \sum m r^2 \text{ (for a system of particles)}$$

$$I = \int r^2 dm \text{ (for a continuous body)}$$

Where: r = perpendicular distance of mass element from the axis, m = mass, dm = elemental mass

2. MOMENT OF INERTIA OF A SOLID CIRCULAR ROD (CYLINDER)

Consider a solid cylinder (circular rod) of mass M , radius R , and length L .

Case 1: Axis along the geometric axis (longitudinal axis)

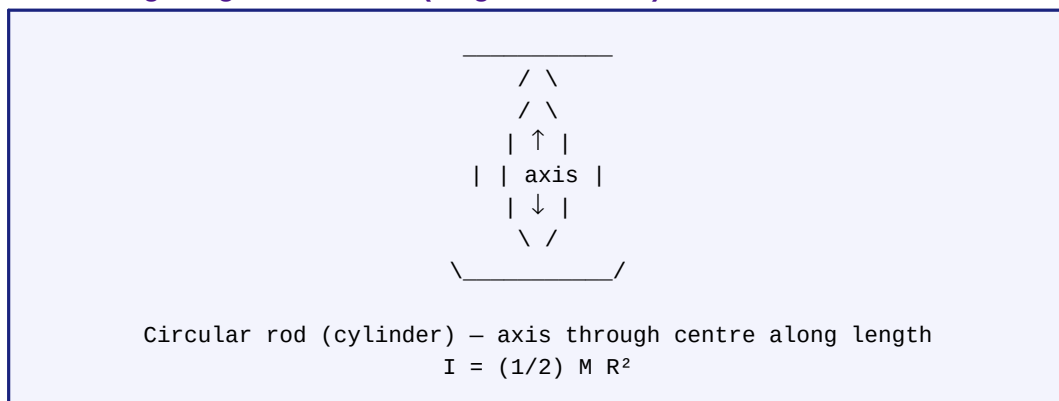


Fig: Cylinder rotating about its own axis (longitudinal)

$$I = (1/2) M R^2$$

Where: M = mass of the rod, R = radius of the circular cross-section

Case 2: Axis perpendicular to the rod and passing through its centre

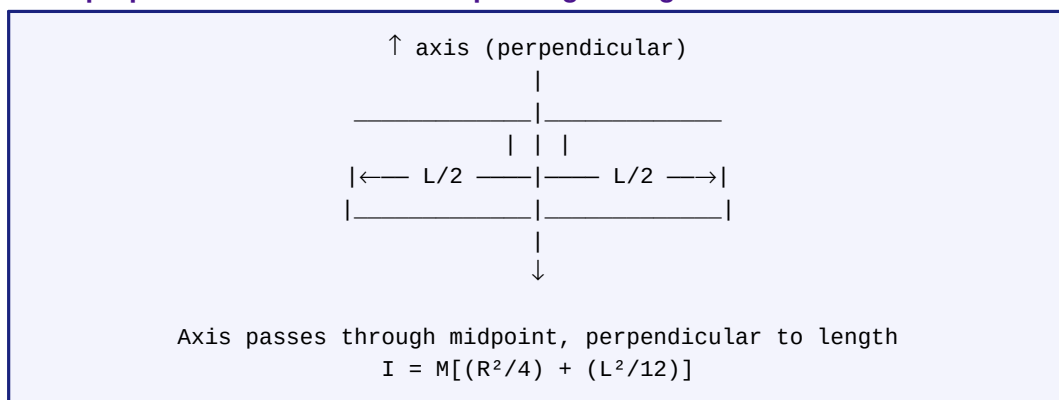


Fig: Cylinder rotating about perpendicular axis through centre

$$I = M[(R^2/4) + (L^2/12)]$$

Where: L = length of the cylinder/rod, R = radius of cross-section

3. SPECIAL CASES FOR A THIN ROD ($R \rightarrow 0$)

If the rod is thin (R is negligible compared to length L), we ignore R :

$$I = ML^2/12 \text{ (axis at centre, perpendicular to rod)}$$

$$I = ML^2/3 \text{ (axis at one end, perpendicular to rod — by parallel axis theorem)}$$

4. PHYSICAL SIGNIFICANCE

- Greater I means more force (torque) is needed for the same angular acceleration.
- $\tau = I \times \alpha$ (torque = moment of inertia $\times$ angular acceleration)
- Used in designing flywheels, shafts, and rotating machinery.

★ **EXAM TIP:** Always write the formula for both axes — along the axis ($I = MR^2/2$) and perpendicular through centre. Write the formula for thin rod as a special case. Draw both diagrams.

Where: ΔD = change in diameter (lateral), D = original diameter, ΔL = change in length (longitudinal), L = original length

The negative sign ensures ν is positive since lateral and longitudinal changes are always in opposite directions.

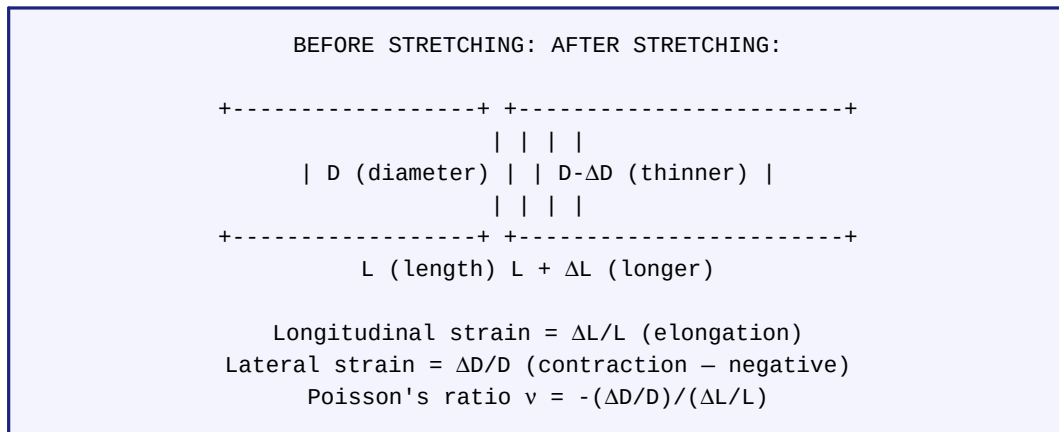


Fig: Poisson's Ratio — Lateral contraction during longitudinal stretch

3. TYPICAL VALUES OF POISSON'S RATIO

Material	Poisson's Ratio (ν)
Steel	0.25 – 0.30
Aluminium	0.33
Rubber	0.48 – 0.50
Cork	≈ 0 (used in bottle stoppers)
Concrete	0.1 – 0.2
Theoretical range	$-1 \leq \nu \leq 0.5$

4. RELATION BETWEEN ELASTIC CONSTANTS

$E = 2G(1 + \nu) \rightarrow$ Relation between E, G (Shear Modulus) and ν

$E = 3K(1 - 2\nu) \rightarrow$ Relation between E, K (Bulk Modulus) and ν

$E = 9KG / (3K + G)$

5. KEY POINTS

- Hooke's Law is valid only within the **elastic limit**.
- Poisson's ratio is always between 0 and 0.5 for most engineering materials.
- $\nu = 0.5$ means incompressible material (like rubber) — volume doesn't change.
- $\nu = 0$ means no lateral deformation (like cork — ideal for bottle stoppers).

★ **EXAM TIP:** Write both Hooke's Law AND Poisson's ratio with formulas and diagrams. The stress-strain curve diagram with Hooke's Law region marked clearly is important. Also mention the range of ν values.

PART A

Question 10: Write a note on Form Girder (I-section Girder).

[13 Marks | Anna University Pattern]

1. INTRODUCTION TO GIRDERS

A **girder** is a large beam used in structural engineering to support heavy loads across long spans (bridges, buildings). The most efficient shape for a girder is the **I-section** (also called I-beam, H-beam, or rolled steel joist — RSJ).

2. BENDING OF BEAMS — THE PROBLEM

When a beam bends under a load:

- The **upper surface** is under **compression** (gets shorter).
- The **lower surface** is under **tension** (gets longer).
- There is a **neutral layer** in the middle where stress = 0.
- The maximum stress is at the top and bottom surfaces (farthest from neutral axis).

Therefore, the material near the top and bottom surfaces carries the most stress. The material near the neutral axis carries very little stress.

3. PRINCIPLE OF I-GIRDER (FORM GIRDER)

In an I-girder, most material is concentrated at the **top and bottom flanges** (where stress is maximum) and very little material is used in the vertical web (where stress is near zero). This makes the I-girder:

- Very **strong** — flanges carry maximum bending stress efficiently.
- Very **light** — minimum material used where stress is negligible.
- Most **economical** — maximum strength-to-weight ratio.

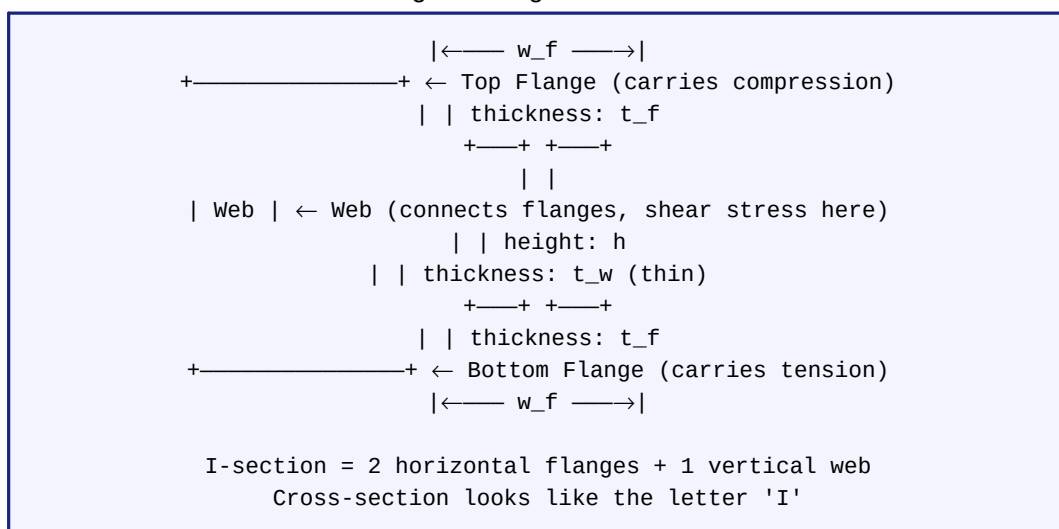


Fig: I-Section Girder (Form Girder) — Cross-sectional view

5. MATHEMATICAL EXPRESSION

The bending moment capacity / section modulus for I-section is much higher than a solid rectangular beam of the same weight:

$$\text{Section Modulus } Z = I / y_{\max}$$

Where I = moment of inertia about the neutral axis, y_{max} = distance from neutral axis to the extreme fibre (top or bottom of flange)

For I-section: $I = \frac{w_f \times d^3}{12} - 2 \times \left[\frac{(w_f - t_w)}{2} \times \frac{h^3}{12} \right]$

The I-section has a higher I value per unit weight, making it more efficient.

6. ADVANTAGES OF I-SECTION GIRDER

- Maximum bending strength with minimum material weight.
- High section modulus → can resist large bending moments.
- Economical — less steel required for the same strength as solid beam.
- Easy to manufacture by rolling process.
- Widely available as standard sections (RSJ — Rolled Steel Joist).

7. DISADVANTAGES

- Weak in torsion (twisting) — a solid beam is better for torsional loads.
- Web may buckle under heavy compressive loads (web buckling).
- Not suitable for loads applied from multiple directions.

8. APPLICATIONS

- Bridges — main longitudinal beams.
- Building floors — carrying floor slabs.
- Cranes and gantry beams.
- Railway tracks — rail sleeper supports.
- Columns in multi-storey buildings.

★ **EXAM TIP:** The key exam point is: 'Material concentrated where stress is maximum (flanges) and minimal where stress is zero (neutral axis) — this is the principle of the I-girder.' Draw the I-section with labels and write advantages.

PART A

Question 11: Briefly explain the efficiency of a Hollow shaft and compare it with a solid shaft of the same material, mass, and length.

[13 Marks | Anna University Pattern]

1. INTRODUCTION TO SHAFTS

A **shaft** is a rotating machine element that transmits power (torque). The efficiency of a shaft depends on how well it resists twisting (torsion) per unit mass.

2. TORSION — BASIC FORMULA

For a shaft subjected to torque T , the torsion formula is:

$$T/J = \tau/r = G\theta/L$$

Where: T = applied torque, J = polar moment of inertia, τ = shear stress, r = outer radius, G = modulus of rigidity, θ = angle of twist, L = length

$$\text{Torque transmitted } T = \tau \times J / r$$

3. POLAR MOMENT OF INERTIA

$$\text{For SOLID shaft (radius } R\text{): } J_s = \pi R^4/2$$

$$\text{For HOLLOW shaft (outer } R, \text{ inner } r\text{): } J_h = \pi(R^4 - r^4)/2$$

4. EFFICIENCY COMPARISON — SAME MASS AND LENGTH

Let the hollow shaft have outer radius R and inner radius r . For **same mass**: cross-sectional area must be equal.

$$\text{Area of solid shaft} = \pi R_s^2$$

$$\text{Area of hollow shaft} = \pi(R^2 - r^2)$$

$$\text{Setting equal: } R_s^2 = R^2 - r^2$$

Comparing torque carrying capacity:

$$T_{\text{hollow}} / T_{\text{solid}} = J_h/R \div J_s/R_s$$

$$= [\pi(R^4 - r^4)/2]/R \div [\pi R_s^4/2]/R_s$$

$$= (R^4 - r^4)/R \times R_s/R_s^4$$

It can be shown mathematically that $T_{\text{hollow}} > T_{\text{solid}}$ for the same mass. This means the hollow shaft is **more efficient**.

5. WHY IS THE HOLLOW SHAFT MORE EFFICIENT?

In a shaft under torsion, the **shear stress is maximum at the outer surface** and **zero at the centre**. The central core contributes very little to torque resistance. By removing the ineffective central core (making it hollow), we:

- Keep the same outer radius (high stress-carrying capacity).
- Reduce mass significantly.
- Improve strength-to-weight ratio.

- Get higher polar moment of inertia per unit mass.

6. COMPARISON TABLE

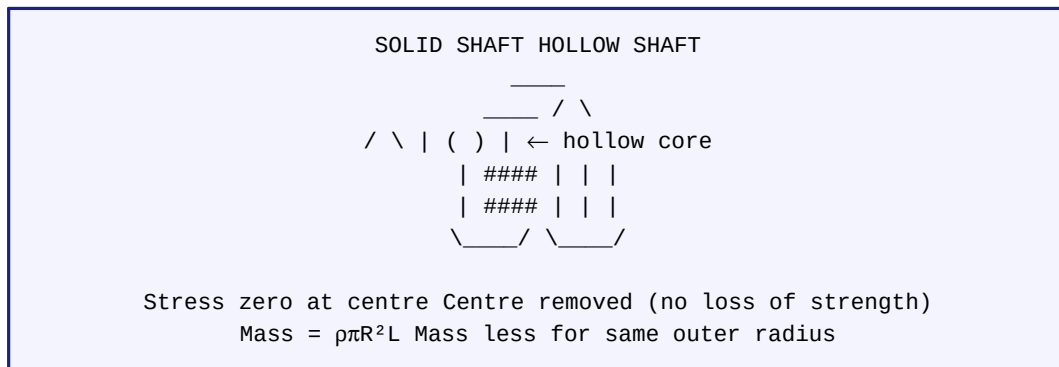


Fig: Cross-section: Solid vs Hollow Shaft

Property	Solid Shaft	Hollow Shaft
Polar moment J	$\pi R^4/2$	$\pi(R^4 - r^4)/2$
Torque capacity (same mass)	Lower	Higher
Weight	More (heavier)	Less (lighter)
Strength-to-weight ratio	Lower	Higher
Material utilization	Poor (centre unused)	Good (no wasted central material)
Manufacturing	Simpler	Slightly complex
Applications	Light-duty shafts	Aircraft, vehicles, high-speed machinery
Cost	Lower initially	More economical in heavy applications

7. QUANTITATIVE RESULT

For a hollow shaft with $r = kR$ (where k = inner/outer radius ratio):

$$\text{Ratio of torques} = T_h/T_s = (1 + k^2) / \sqrt{1 - k^2} > 1$$

Since this ratio is always greater than 1, hollow shaft always transmits MORE torque per unit mass than solid shaft.

★ EXAM TIP: The key phrase for exam: 'Shear stress is zero at centre and maximum at surface. A hollow shaft removes the zero-stress core, improving efficiency for same mass.' Write the formula for J in both cases.

$$v_t = 2r^2(\rho - \sigma)g / 9\eta$$

Where: r = radius of sphere, ρ = density of sphere, σ = density of fluid, $g = 9.8 \text{ m/s}^2$, η = coefficient of viscosity

5. SIGNIFICANCE OF VISCOUS DRAG

- **Engineering:** Viscous drag in pipes determines pumping power required for fluid transport (oil pipelines, water supply).
- **Aerodynamics:** Viscous drag on aircraft determines fuel efficiency and speed (skin friction drag).
- **Automobiles:** Drag-reducing vehicle shapes reduce viscous air resistance and improve mileage.
- **Lubrication:** Viscous oil film in bearings reduces wear; viscosity must be optimum.
- **Medicine:** Blood viscosity affects blood flow in arteries and veins.
- **Sports:** Swimmers, cyclists, and runners wear streamlined clothing to reduce viscous drag.
- **Measurement:** Stokes' Law is used in viscometers to measure viscosity of liquids.
- **Sedimentation:** Smaller particles settle slower due to viscous drag — used in centrifuges.

6. TYPES OF FLOW

- **Laminar (viscous) flow:** Smooth, parallel layers. Low velocity. $Re < 2000$.
- **Turbulent flow:** Chaotic, irregular. High velocity. $Re > 4000$.
- **Reynolds Number $Re = \rho v D / \eta$** — determines whether flow is laminar or turbulent.

★ **EXAM TIP:** Viscous drag significance means: write about Stokes' Law, terminal velocity formula, and then list 5-6 real-life applications. This scores well.

PART A

Question 13: Distinguish Solid and Hollow shafts with suitable relationships based on twist.

[13 Marks | Anna University Pattern]

1. TORSION EQUATION (FUNDAMENTAL)

The torsion equation relating torque, shear stress, and angle of twist is:

$$T/J = \tau/R = G\theta/L$$

Where: T = applied torque (N·m), J = polar moment of inertia (m⁴), τ = shear stress (N/m²), R = outer radius (m), G = modulus of rigidity (N/m²), θ = angle of twist (rad), L = length of shaft (m)

2. RELATIONSHIPS FOR SOLID SHAFT

Polar moment of inertia: $J_s = \pi D^4/32 = \pi R^4/2$

Maximum shear stress: $\tau_s = 16T / (\pi D^3) = TR / J_s$

Angle of twist: $\theta_s = TL / (G \times J_s) = 32TL / (G\pi D^4)$

Power transmitted: $P = 2\pi NT / 60$

3. RELATIONSHIPS FOR HOLLOW SHAFT

Let outer diameter = D, inner diameter = d, $k = d/D$

Polar moment of inertia: $J_h = \pi(D^4 - d^4)/32$

Maximum shear stress: $\tau_h = 16TD / [\pi(D^4 - d^4)]$

Angle of twist: $\theta_h = 32TL / [G\pi(D^4 - d^4)]$

4. COMPARISON TABLE — SOLID vs HOLLOW SHAFT

Parameter	Solid Shaft	Hollow Shaft
Polar Moment J	$\pi D^4/32 = \pi R^4/2$	$\pi(D^4 - d^4)/32 = \pi(R^4 - r^4)/2$
Max. Shear Stress τ	$16T/\pi D^3$	$16TD/\pi(D^4 - d^4)$
Angle of Twist θ	$32TL/G\pi D^4$	$32TL/G\pi(D^4 - d^4)$
Torque Capacity (same mass)	Lower	Higher by factor $(1+k^2)/\sqrt{1-k^2}$
Strength/Weight	Lower	Higher
Weight	More	Less (lighter)
Application	Light duty, low-speed machinery	Aircraft, ship propellers, drilling rigs

5. PROOF: HOLLOW SHAFT IS STRONGER PER UNIT WEIGHT

For same mass and material: Area of solid = Area of hollow

$$\pi R_s^2 = \pi(R^2 - r^2) \rightarrow R_s^2 = R^2 - r^2$$

Torque ratio (for same max. stress τ):

$$\begin{aligned}
 T_h / T_s &= (J_h / R) / (J_s / R_s) \\
 &= [(R^4 - r^4) / R] / [R_s^4 / R_s] \\
 &= (R^2 + r^2)(R^2 - r^2) / R \times R_s / R_s^4 \\
 &= (R^2 + r^2)(R_s^2) / R_s \times (1/R) \text{ [since } R^2 - r^2 = R_s^2 \text{]} \\
 T_h / T_s &= (R^2 + r^2) / R_s^2 \times (R_s / R) > 1
 \end{aligned}$$

This proves that **hollow shaft transmits more torque for same mass.**

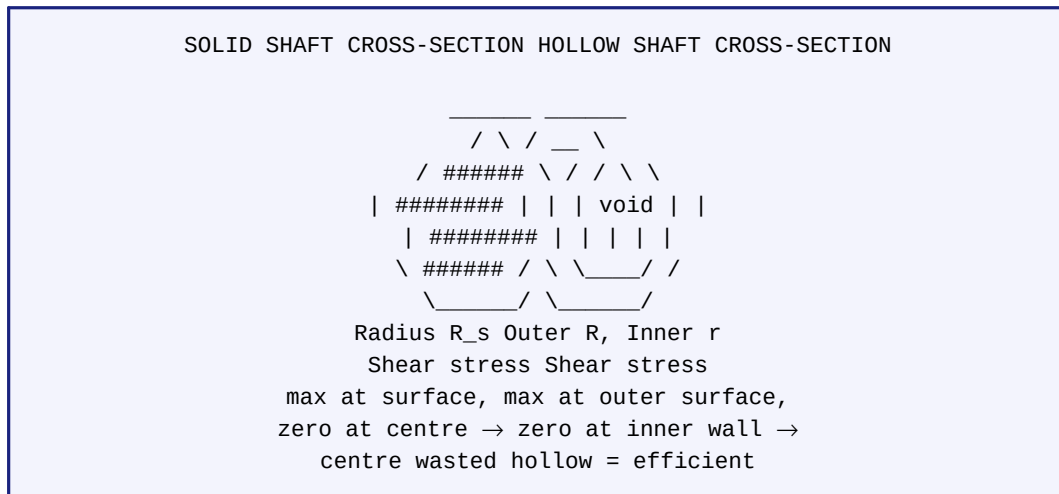


Fig: Solid vs Hollow shaft cross-section shear stress distribution

★ **EXAM TIP: Write ALL THREE relationships: J , τ , and θ for BOTH shafts side by side. The comparison table is essential. Also add the torque ratio proof.**

PART B

Question 1: Explain in detail the Atomic Packing Factor (APF) in HCP and FCC structures with a neat diagram.

[13 Marks | Anna University Pattern]

1. DEFINITION OF ATOMIC PACKING FACTOR (APF)

Atomic Packing Factor (APF) is defined as the fraction of the total volume of the unit cell that is actually occupied by atoms. It tells us how efficiently atoms are packed in a crystal structure.

APF = Volume occupied by atoms in unit cell / Total volume of unit cell

$$\text{APF} = (n \times (4/3)\pi r^3) / V_{\text{cell}}$$

Where: n = number of atoms per unit cell, r = atomic radius, V_{cell} = volume of unit cell

APF is always less than 1 (less than 100%). Higher APF = denser packing.

2. APF OF FCC (FACE CENTRED CUBIC) STRUCTURE

Structure of FCC:

- Atoms at all 8 corners of the cube.
- Atoms at the centre of all 6 faces.
- Each corner atom shared by 8 unit cells → contributes 1/8.
- Each face atom shared by 2 unit cells → contributes 1/2.

$$n = 8 \times (1/8) + 6 \times (1/2) = 1 + 3 = 4 \text{ atoms per unit cell}$$

Relation between r and a for FCC:

Atoms touch along the face diagonal:

$$\text{Face diagonal} = 4r = a\sqrt{2}$$

$$\therefore r = a\sqrt{2} / 4 \text{ or } a = 2\sqrt{2} r$$

Calculation of APF for FCC:

$$\text{Volume of } n=4 \text{ atoms} = 4 \times (4/3)\pi r^3 = (16/3)\pi r^3$$

$$\text{Volume of unit cell} = a^3 = (2\sqrt{2} r)^3 = 16\sqrt{2} r^3$$

$$\text{APF} = (16\pi r^3/3) / (16\sqrt{2} r^3)$$

$$\text{APF} = \pi / (3\sqrt{2}) = 3.1416 / (3 \times 1.4142) = 3.1416 / 4.2426$$

$$\text{APF (FCC)} = 0.7405 = 74.05\%$$

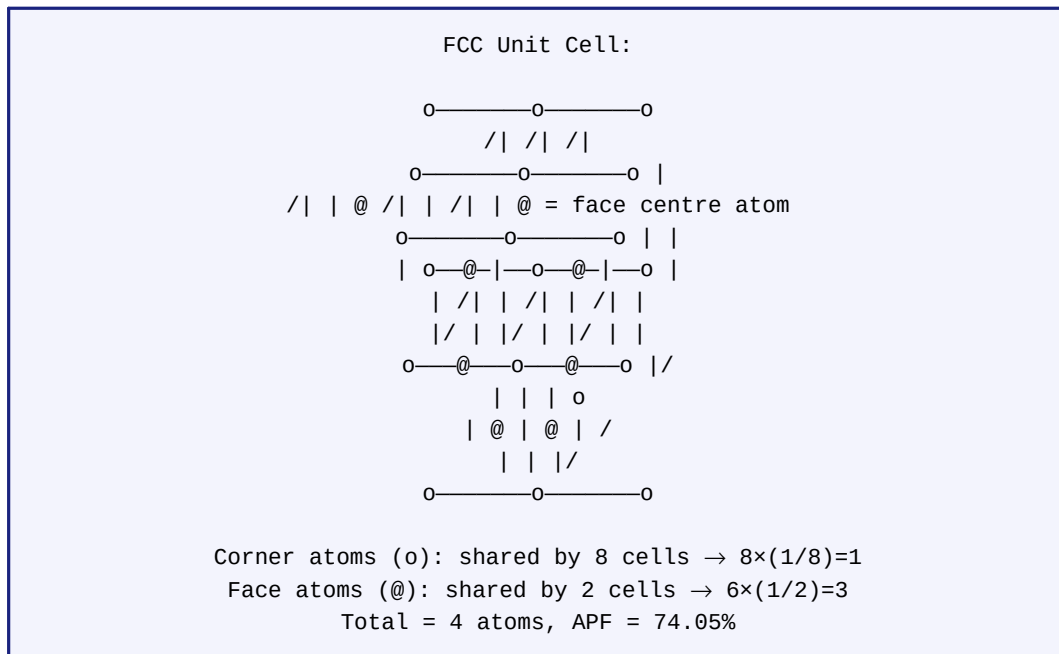


Fig: FCC Structure — 4 atoms per unit cell

3. APF OF HCP (HEXAGONAL CLOSE-PACKED) STRUCTURE

Structure of HCP:

- Two hexagonal close-packed layers (top and bottom) + one middle layer.
- Top and bottom layers each have 6 corner atoms + 1 face centre atom.
- Middle layer has 3 atoms completely inside the cell.

Number of atoms per HCP unit cell:

- 12 corner atoms (top 6 + bottom 6): each shared by 6 cells $\rightarrow 12 \times (1/6) = 2$
- 2 face centre atoms (top + bottom): each shared by 2 cells $\rightarrow 2 \times (1/2) = 1$
- 3 middle layer atoms: completely inside $\rightarrow 3 \times 1 = 3$

Total $n = 2 + 1 + 3 = 6$ atoms per HCP unit cell

Geometry of HCP unit cell:

Lattice parameter: $a = 2r$ (atoms touch along base edge)

Height of unit cell: $c = 2\sqrt{2/3} \times a = 1.633 \times a$

Volume of HCP cell: $V = 3\sqrt{3}/2 \times a^2 \times c = 3\sqrt{3}/2 \times a^2 \times 1.633a$

$V = 3\sqrt{3}/2 \times 1.633 \times a^3 = 3\sqrt{2} \times a^3 = 3\sqrt{2} \times (2r)^3 = 24\sqrt{2} r^3$

Calculation of APF for HCP:

Volume of 6 atoms $= 6 \times (4/3)\pi r^3 = 8\pi r^3$

$\text{APF} = 8\pi r^3 / 24\sqrt{2} r^3 = 8\pi / 24\sqrt{2} = \pi / 3\sqrt{2}$

$\text{APF (HCP)} = 0.7405 = 74.05\%$

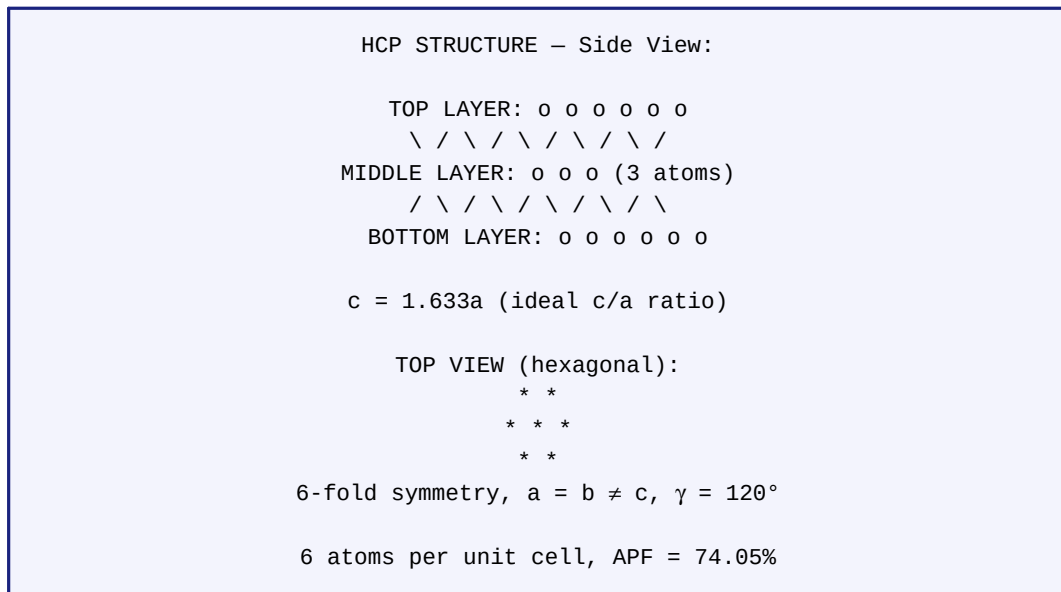


Fig: HCP Structure — 6 atoms per unit cell

4. COMPARISON: FCC vs HCP

Property	FCC	HCP
Crystal System	Cubic	Hexagonal
Atoms/unit cell	4	6
Coordination Number	12	12
APF	74.05%	74.05%
Stacking Sequence	ABCABC...	ABABAB...
Examples	Cu, Al, Ni, Au, Ag	Zn, Mg, Ti, Co, Cd
Slip Systems	12 (more ductile)	3 (less ductile)

Important note: Both FCC and HCP have the same APF (74.05%), which is the highest possible for equal-sphere packing. Both are called **close-packed structures**. The difference is only in stacking sequence.

5. COMPARISON WITH OTHER STRUCTURES

Structure	APF	Coordination No.	Examples
Simple Cubic (SC)	52.36%	6	Polonium (only element)
BCC	68.02%	8	Fe(α), W, Cr, Mo
FCC	74.05%	12	Cu, Al, Ni, Au
HCP	74.05%	12	Zn, Mg, Ti
Diamond Cubic	34%	4	C (diamond), Si, Ge

★ **EXAM TIP:** Calculate APF step by step: (1) find n, (2) relate r to a, (3) find volume of atoms and cell, (4) divide. The APF of FCC = HCP = 74.05% — this is a very frequently asked fact. The coordination number of both = 12.

PART B

Question 2: Explain in detail the growth of layers by VPE and MBE techniques.

[13 Marks | Anna University Pattern]

1. INTRODUCTION TO THIN FILM GROWTH

Modern semiconductor devices (transistors, LEDs, solar cells) are made by depositing very thin crystal layers on a substrate. The two most important techniques are:

- **VPE** — Vapour Phase Epitaxy
- **MBE** — Molecular Beam Epitaxy

Epitaxy means growing a single-crystal thin film on a crystalline substrate such that the film adopts the crystal structure of the substrate.

2. VPE — VAPOUR PHASE EPITAXY

Principle:

In VPE, gaseous (vapour) precursor chemicals are introduced into a heated reaction chamber. These gases react at the hot substrate surface and deposit a thin crystalline layer. The rest of the gases (by-products) are removed.

Types of VPE:

- **HVPE (Hydride VPE):** Uses hydride gases (e.g., AsH₃, PH₃).
- **MOVPE / MOCVD (Metal-Organic VPE):** Uses metal-organic compounds like TMGa (trimethylgallium), TMAI. Most widely used for commercial LEDs.
- **CVD (Chemical Vapour Deposition):** General form of VPE.

Process Steps in VPE:

Step 1: Substrate (e.g., GaAs, Si) is placed in the reaction chamber.

Step 2: Chamber is heated to 600–1000°C.

Step 3: Precursor gases are flowed over the substrate surface.

Step 4: Gases react chemically at the hot surface → atoms deposit and crystallize.

Step 5: By-products (e.g., HCl, CH₄) are carried away by the carrier gas (H₂ or N₂).

Step 6: Layer-by-layer crystalline film grows on the substrate.

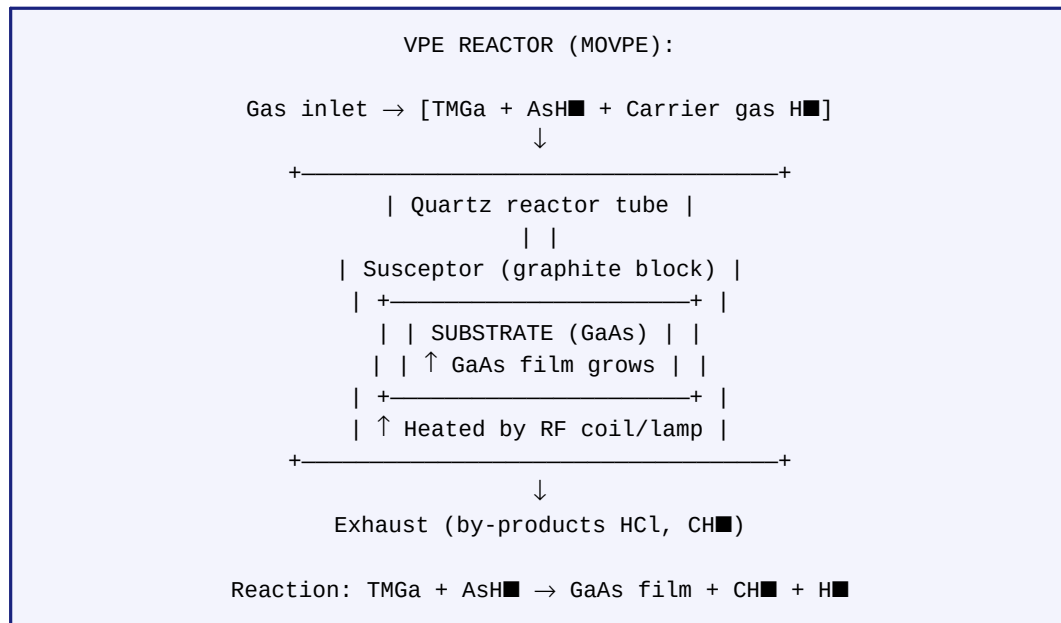


Fig: VPE (MOVPE) Reactor — Schematic Diagram

Advantages of VPE:

- High growth rate (1–10 $\mu\text{m/hr}$) — suitable for thick layers.
- Large area deposition possible (batch processing).
- Can grow compound semiconductors (GaAs, InP, GaN, AlGaAs).
- Compatible with commercial mass production.
- Precise control over composition by adjusting gas flow ratios.

Disadvantages of VPE:

- High temperature can cause unintended diffusion or defects.
- Toxic and flammable precursor gases (AsH₃, PH₃, H₂).
- Not suitable for extremely thin (monolayer) films.
- Expensive gas delivery and exhaust treatment systems.

3. MBE — MOLECULAR BEAM EPITAXY

Principle:

In MBE, ultra-pure solid source materials are heated in separate **effusion cells** inside a ultra-high vacuum (UHV) chamber (pressure $\sim 10^{-11}$ torr). The evaporated atoms or molecules travel as **molecular beams** and condense on a heated substrate, growing a crystalline layer one monolayer (atomic layer) at a time.

Construction of MBE System:

- **Effusion Cells (Knudsen cells):** Contain source materials (Ga, As, Al, In, Si, etc.). Heated electrically. Each cell has a shutter.
- **Substrate Holder:** Rotates to ensure uniform deposition. Heated to 500–700°C.
- **UHV Chamber:** Maintained at 10^{-11} to 10^{-10} torr. No gas-phase reactions.
- **RHEED Gun:** Reflection High Energy Electron Diffraction — monitors growth in real time.
- **Cooling shroud:** Liquid nitrogen-cooled walls capture stray atoms.

Process Steps in MBE:

Step 1: Substrate is loaded in UHV chamber and cleaned.

Step 2: Effusion cells are heated to generate molecular beams.

Step 3: Shutters of cells are opened — beams of atoms hit the substrate.

Step 4: Atoms condense and crystallize epitaxially on the substrate.

Step 5: Shutters are used to control thickness with monolayer precision.

Step 6: RHEED monitors growth to detect each monolayer completion.

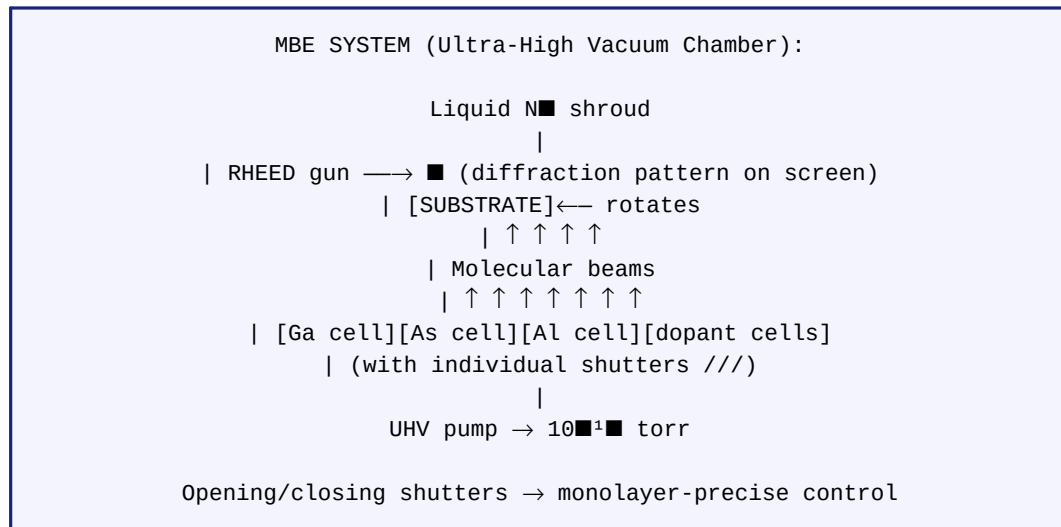


Fig: MBE System — Schematic Diagram

Advantages of MBE:

- Atomic layer precision — can grow films just 1–2 monolayers thick.
- Ultra-pure films — UHV prevents contamination.
- Real-time monitoring with RHEED during growth.
- Sharp interfaces between layers (quantum well structures).
- Can grow complex heterostructures (GaAs/AlGaAs quantum wells).
- Excellent control of doping profiles.

Disadvantages of MBE:

- Very slow growth rate (0.1–1 μm/hr) — not suitable for thick layers.
- Extremely expensive equipment and maintenance.
- Small deposition area — not easy to scale up for mass production.
- Source materials replenishment requires breaking vacuum.

4. COMPARISON: VPE vs MBE

Feature	VPE (MOVPE)	MBE
Principle	Chemical reaction of gas-phase precursors	Evaporation + physical deposition in UHV
Pressure	Atmospheric or low pressure	Ultra-High Vacuum 10 ⁻¹⁰ to 10 ⁻¹¹ torr
Growth rate	Fast: 1–10 μm/hr	Slow: 0.1–1 μm/hr
Thickness control	Good (nm range)	Excellent (monolayer precision)
Interface sharpness	Good	Excellent (atomic scale)
Film purity	High	Ultra-high
Monitoring	Limited in-situ	Real-time RHEED

Cost	Lower	Very high
Scale-up	Easy	Difficult
Applications	LED, laser diodes, solar cells (GaN, GaAs)	Quantum wells, quantum dots, HEMTs, research devices

5. APPLICATIONS

- **VPE/MOVPE:** Manufacturing of GaN LEDs (blue/white LEDs), laser diodes, GaAs solar cells for satellites.
- **MBE:** Quantum well lasers, High Electron Mobility Transistors (HEMTs), quantum dot structures, research-grade semiconductor wafers.
- Both techniques are essential for modern semiconductor industry.

★ **EXAM TIP:** In the exam, draw both reactor diagrams clearly. Write the comparison table. Key differences: MBE = UHV + monolayer precision + slow. VPE = gas reactions + fast + commercial scale. Both used for compound semiconductors.

PART B

Question 3: Derive an expression for the Bending Moment of a beam. Hence derive an expression for a beam loaded at both ends and has elevation at the centre of the beam.

[13 Marks | Anna University Pattern]

1. INTRODUCTION TO BENDING OF BEAMS

When a beam is loaded transversely, it bends. This bending produces internal **stresses** and **strains** within the beam. The bending moment is the moment of the internal forces acting on a cross-section of the beam.

2. ASSUMPTIONS IN BENDING THEORY

- The beam is initially straight and has uniform cross-section.
- Material is homogeneous, isotropic and obeys Hooke's Law.
- Plane sections remain plane after bending (Euler-Bernoulli assumption).
- The beam bends in the plane of loading.
- Deflections are small compared to beam dimensions.

3. THEORY OF SIMPLE BENDING — DERIVATION

Consider a beam bent into a circular arc. Let:

- R = radius of curvature of the neutral layer
- y = distance of a layer from the neutral axis
- ϵ = strain in the layer
- σ = stress in the layer
- E = Young's modulus

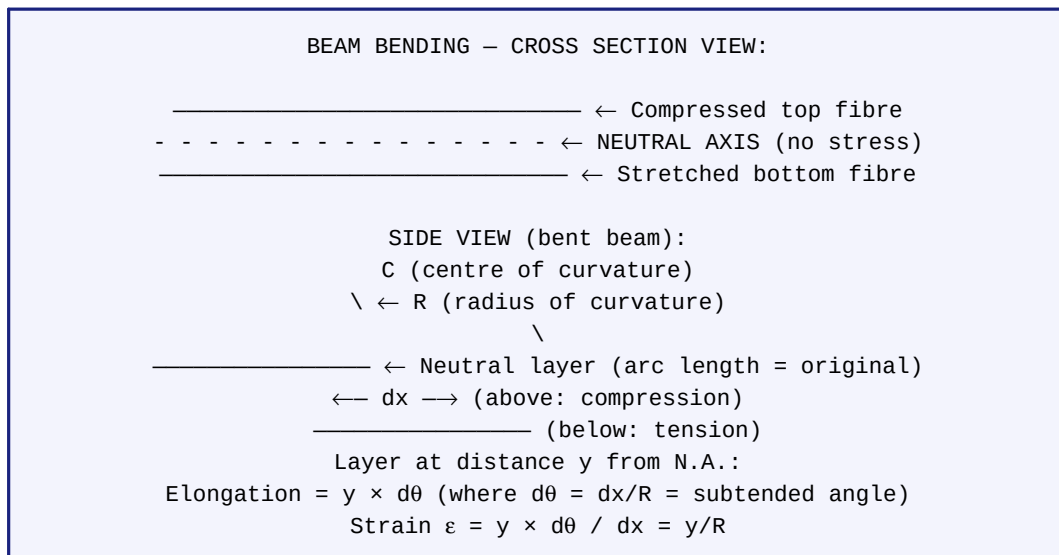


Fig: Bending of beam — Neutral axis and strain distribution

Step-by-step derivation:

From the geometry:

$$\text{Strain in layer at distance } y = \epsilon = y/R$$

By Hooke's Law: $\sigma = E \times \varepsilon = Ey/R$

$$\therefore \sigma/y = E/R \dots (1)$$

The stress σ on an elemental area dA at distance y produces a force:

$$dF = \sigma \times dA = (Ey/R) \times dA$$

$$\text{Moment of this force about N.A.} = dF \times y = (E/R) \times y^2 \times dA$$

Total bending moment $M = \int (E/R) \times y^2 \times dA$:

$$M = (E/R) \times \int y^2 dA = (E/R) \times I$$

$$\therefore M/I = E/R \dots (2)$$

From (1) and (2):

$$M/I = \sigma/y = E/R$$

This is the **Bending Equation** (Flexure Formula).

4. EXPLANATION OF TERMS IN BENDING EQUATION: $M/I = \sigma/y = E/R$

Symbol	Meaning	Unit
M	Bending Moment at the cross-section	N·m
I	Second Moment of Area (Moment of Inertia) about the neutral axis	m ⁴
σ	Bending stress at distance y from N.A.	N/m ² (Pa)
y	Distance from neutral axis to the fibre under consideration	m
E	Young's Modulus of the material	N/m ² (Pa)
R	Radius of curvature of neutral layer	m

5. BEAM LOADED AT BOTH ENDS WITH ELEVATION AT CENTRE (Non-Uniform Bending)

This setup is used to find Young's Modulus of a material experimentally.

Setup / Description:

- A beam of length L is supported at both ends (supports at A and B).
- A point load W is applied at the centre C.
- The centre of the beam deflects downward by δ (elevation = depression at centre).
- By symmetry, each support reaction = $W/2$.

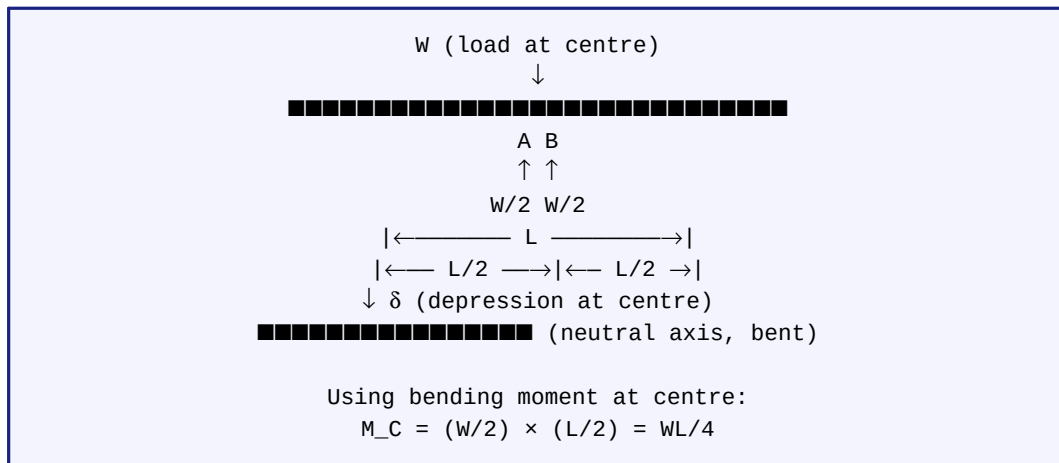


Fig: Simply Supported Beam with Central Point Load

Derivation of depression at centre:

The bending moment at a distance x from support A (for $0 \leq x \leq L/2$):

$$M(x) = (W/2) \times x$$

Using the bending moment $M = EI(d^2y/dx^2)$ and integrating:

$$EI (d^2y/dx^2) = (W/2) \times x$$

$$EI (dy/dx) = (W/2)(x^2/2) + C_1 = Wx^2/4 + C_1$$

$$EI \times y = Wx^3/12 + C_1x + C_2$$

Applying boundary conditions:

- At $x = 0$ (support A): $y = 0 \rightarrow C_2 = 0$
- At $x = L/2$ (centre): $dy/dx = 0$ (maximum deflection) $\rightarrow C_1 = -WL^2/16$

$$EI \times y = Wx^3/12 - (WL^2/16)x$$

Maximum deflection at $x = L/2$:

$$EI \times \delta = W(L/2)^3/12 - (WL^2/16)(L/2)$$

$$EI \times \delta = WL^3/96 - WL^3/32 = WL^3/96 - 3WL^3/96 = -2WL^3/96 = -WL^3/48$$

$$\therefore \delta = WL^3 / (48EI) \leftarrow \text{Maximum depression at centre}$$

Expression for Young's Modulus:

$$E = WL^3 / (48 \times I \times \delta)$$

For a rectangular beam of breadth b and depth d : $I = bd^3/12$

$$\therefore E = WL^3 / (48 \times (bd^3/12) \times \delta) = WL^3 / (4bd^3\delta)$$

$$E = WL^3 / (4bd^3\delta)$$

Where: W = applied load, L = length between supports, b = breadth, d = depth (thickness) of beam, δ = depression (measured by travelling microscope)

6. APPLICATIONS OF BENDING THEORY

- Design of beams and girders in buildings and bridges.
- Experimental determination of Young's Modulus (Uniform bending, Non-uniform bending).
- Design of springs, leaf springs in automobiles.
- Analysis of aircraft wings and fuselage.

★ EXAM TIP: The bending equation $M/I = \sigma/y = E/R$ must be fully derived with the diagram. For non-uniform bending, derive the formula $\delta = WL^3/48EI$ step by step. Finally write $E = WL^3/(4bd^3\delta)$ as the experimental formula.

PART B

Question 4: Derive an expression for the determination of Young's Modulus by Cantilever and hence Non-Uniform bending of a beam.

[13 Marks | Anna University Pattern]

1. INTRODUCTION TO CANTILEVER

A **cantilever** is a beam that is **fixed (clamped) at one end** and **free at the other end**. When a load is applied at the free end, the beam bends. This setup is used to determine Young's Modulus.

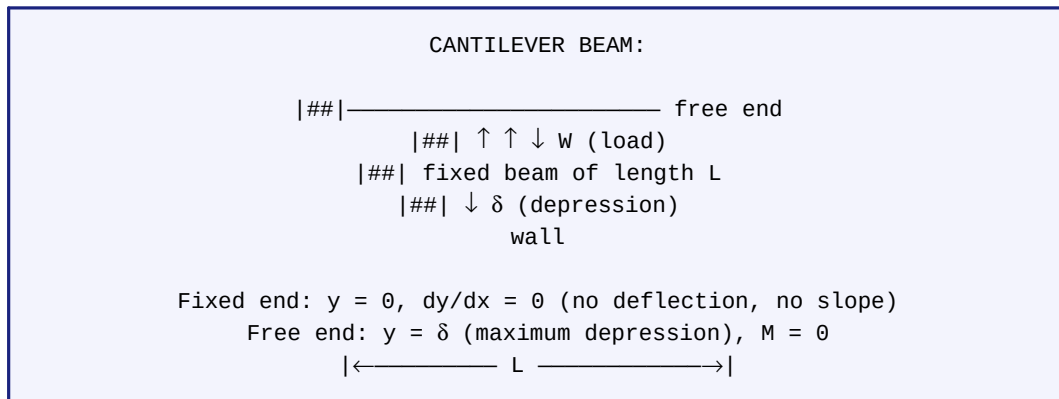


Fig: Cantilever — Fixed at one end, loaded at free end

2. BENDING MOMENT IN A CANTILEVER

Take origin at the free end. At a distance x from the free end, the bending moment is:

$$M(x) = -W \times x \text{ (negative because it causes hogging)}$$

Using the bending equation:

$$EI (d^2y/dx^2) = M = -Wx$$

(Here x is measured from the FREE end toward the fixed end)

3. DERIVATION OF DEPRESSION AT FREE END

Integrating the bending equation:

$$EI (dy/dx) = -Wx^2/2 + C_1$$

Boundary condition: At fixed end ($x = L$), $dy/dx = 0$ (no slope):

$$0 = -WL^2/2 + C_1 \rightarrow C_1 = WL^2/2$$

$$\therefore EI (dy/dx) = -Wx^2/2 + WL^2/2$$

Integrating again:

$$EI \times y = -Wx^3/6 + WL^2x/2 + C_2$$

Boundary condition: At fixed end ($x = L$), $y = 0$:

$$0 = -WL^3/6 + WL^3/2 + C_2$$

$$C_2 = WL^3/6 - WL^3/2 = -WL^3/3$$

$$\therefore EI \times y = -Wx^3/6 + WL^2x/2 - WL^3/3$$

Depression at FREE end ($x = 0$):

$$EI \times \delta = 0 + 0 - WL^3/3$$

$$\delta = WL^3 / (3EI)$$

Young's Modulus from Cantilever:

$$E = WL^3 / (3 \times I \times \delta)$$

For rectangular cross-section beam (breadth b, depth d): $I = bd^3/12$

$$E = WL^3 / (3 \times bd^3/12 \times \delta) = 4WL^3 / (bd^3\delta)$$

$$E = 4WL^3 / (bd^3\delta)$$

Where: W = applied load at free end (N), L = length of cantilever (m), b = breadth of beam (m), d = depth of beam (m), δ = depression (m)

4. NON-UNIFORM BENDING — OVERVIEW

In **non-uniform bending**, the bending moment varies along the length of the beam. The most common case is a simply supported beam with a central load (already derived: $\delta = WL^3/48EI$).

The key formula for Young's Modulus in non-uniform bending (central loading):

$$E = WL^3 / (48 \times I \times \delta)$$

$$\text{For rectangular section: } E = WL^3 / (4bd^3\delta)$$

5. COMPARISON — CANTILEVER vs NON-UNIFORM BENDING

Feature	Cantilever	Non-Uniform Bending (Simply Supported + Central Load)
Support	Fixed at one end, free at other	Supported at both ends
Load position	Free end	Centre
Max. deflection location	Free end	Centre of beam
Formula for δ	$WL^3/3EI$	$WL^3/48EI$
Formula for E	$E = 4WL^3/(bd^3\delta)$	$E = WL^3/(4bd^3\delta)$
Bending moment	Non-uniform (varies linearly)	Non-uniform (triangular BMD)
Stress at fixed end	Maximum	—
Applications	Diving boards, balconies	Beams, bridges

6. EXPERIMENTAL MEASUREMENT

In the laboratory:

- Beam of known dimensions (L, b, d) placed as cantilever or on supports.
- Known weight W placed at free end (cantilever) or centre (non-uniform).
- Depression δ measured using a **travelling microscope** or dial gauge.
- Young's Modulus calculated using the appropriate formula.

★ **EXAM TIP:** Write both derivations clearly. The key result: $\delta = WL^3/3EI$ (cantilever) vs $\delta = WL^3/48EI$ (simply supported). From these, derive E. Draw the diagrams showing the load configuration and deflected shape.

PART B

Question 5: Define Torsional Oscillations and derive the expression for Couple per unit twist.

[13 Marks | Anna University Pattern]

1. INTRODUCTION — TORSION OF A WIRE

When a cylindrical wire or rod is twisted, it resists the twisting due to internal shear forces. When released, the wire exerts a **restoring couple** and the body undergoes oscillatory rotational motion — called **Torsional Oscillations**.

2. DEFINITION — TORSIONAL OSCILLATIONS

When a body suspended from a wire is twisted through an angle and released, it oscillates angularly (rotates back and forth) about the wire axis. These oscillations are called **Torsional (Vibrational) Oscillations**.

The restoring torque is proportional to the angle of twist: $\tau = -C\theta$, making them simple harmonic oscillations.

3. DERIVATION OF COUPLE PER UNIT TWIST (C)

Consider a cylindrical wire of:

- Length L , radius r
- Modulus of rigidity η (or G)
- Twisted at one end by an angle ϕ (phi)

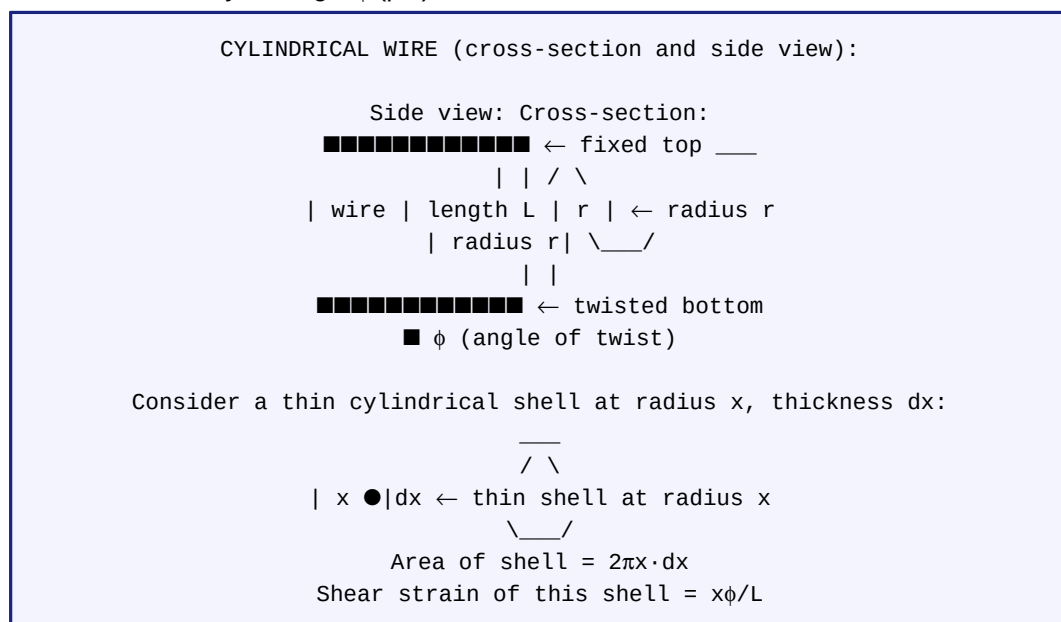


Fig: Cylindrical wire under torsion — elemental shell analysis

Step 1 — Shear strain and shear stress in elemental shell:

For a cylindrical shell at radius x , thickness dx , the angle of twist = ϕ :

Shear strain $\gamma = x \times \phi / L$

Shear stress $\tau = \eta \times \gamma = \eta \times x\phi / L$

(Where η = modulus of rigidity)

Step 2 — Tangential shear force on elemental ring:

$$\text{Area of ring} = 2\pi x \times dx$$

$$\text{Shear force on ring } dF = \tau \times dA = (\eta x \phi / L) \times 2\pi x \, dx = (2\pi \eta \phi / L) \times x^2 \, dx$$

Step 3 — Couple (torque) due to this elemental ring:

$$dC = dF \times x = (2\pi \eta \phi / L) \times x^3 \, dx$$

Step 4 — Total couple (integrate from 0 to r):

$$C_{\text{total}} = \int_0^r (2\pi \eta \phi / L) \times x^3 \, dx$$

$$C_{\text{total}} = (2\pi \eta \phi / L) \times [x^4 / 4]_0^r$$

$$C_{\text{total}} = (2\pi \eta \phi / L) \times (r^4 / 4)$$

$$C_{\text{total}} = \pi \eta \phi r^4 / (2L)$$

Step 5 — Couple per unit twist (C):

Setting $\phi = 1$ radian (unit twist):

$$C = \pi \eta r^4 / (2L)$$

This is the **Couple per unit twist** (also called torsional rigidity of the wire).

In terms of diameter $D = 2r$:

$$C = \pi \eta r^4 / (2L) = \pi \eta D^4 / (32L)$$

4. PERIOD OF TORSIONAL OSCILLATIONS

For a disc of moment of inertia I suspended by the wire:

Restoring torque $\tau = -C \times \theta$ (proportional to angle, so SHM)

By Newton's 2nd law for rotation: $I \times d^2\theta/dt^2 = -C\theta$

$$d^2\theta/dt^2 + (C/I)\theta = 0 \rightarrow \text{SHM with } \omega^2 = C/I$$

$$\text{Time period } T = 2\pi\sqrt{I/C} = 2\pi\sqrt{2IL / \pi \eta r^4}$$

5. USES OF COUPLE PER UNIT TWIST

- Used in **torsional pendulum** to measure moment of inertia of objects.
- Used to find **modulus of rigidity (η)** of wire materials.
- Design of torsion springs in watches, galvanometers.
- Analysis of power transmission shafts and propeller shafts.

6. MODULUS OF RIGIDITY FROM TORSIONAL OSCILLATION

Combining $C = \pi \eta r^4 / 2L$ and $T = 2\pi\sqrt{I/C}$:

$$\eta = 8\pi IL / (r^4 T^2) = 8\pi IL / (r^4 T^2)$$

This is the experimental formula for modulus of rigidity.

★ **EXAM TIP:** The derivation of $C = \pi \eta r^4 / 2L$ is the **MOST IMPORTANT** part — show all 5 steps. Also write $T = 2\pi\sqrt{I/C}$ and the modulus of rigidity formula. The diagram of elemental shell is essential for marks.

QUICK REVISION — KEY FORMULAE SHEET

Concept / Law	Formula
Crystal Systems	7 systems, 14 Bravais lattices
Bragg's Law	$2d \sin\theta = n\lambda$
Frenkel defect concentration	$n = \sqrt{(NN_i)} \exp(-E_f/2kT)$
Schottky defect concentration	$n = N \exp(-E_s/2kT)$
APF — FCC & HCP	74.05% (both same)
APF — BCC	68.02%
Bending Equation	$M/I = \sigma/y = E/R$
Depression — Cantilever	$\delta = WL^3/(3EI)$
Depression — Non-uniform (central load)	$\delta = WL^3/(48EI)$
Young's Modulus (cantilever)	$E = 4WL^3/(bd^3\delta)$
Young's Modulus (non-uniform)	$E = WL^3/(4bd^3\delta)$
Couple per unit twist	$C = \pi\eta r^4/(2L) = \pi\eta D^4/(32L)$
Torsional oscillation period	$T = 2\pi\sqrt{I/C}$
Modulus of rigidity	$\eta = 8\pi IL/(r^4T^2)$
Polar MI — Solid shaft	$J = \pi D^4/32 = \pi R^4/2$
Polar MI — Hollow shaft	$J = \pi(D^4 - d^4)/32$
Torsion equation	$T/J = \tau/R = G\theta/L$
Stokes viscous drag	$F = 6\pi\eta rv$
Terminal velocity	$vt = 2r^2(\rho - \sigma)g/(9\eta)$
Hooke's Law	$\sigma = E\epsilon$
Poisson's ratio	$\nu = -(\Delta D/D)/(\Delta L/L)$
MI thin rod — centre	$I = ML^2/12$
MI thin rod — end	$I = ML^2/3$
MI cylinder — own axis	$I = MR^2/2$

All the best for your Anna University Examinations!
 Study smart, write complete answers, draw neat diagrams, and score maximum marks.